

The Question We Asked

An ECG signal is normally a 1D waveform. We converted these continuous signals into simple 2D pictures (line plots) to treat this as a standard image classification task.

- **The Core Premise:** If we already turn the signal into a picture, do we really need a complex, heavy deep learning model to read it?
- **The Scale:** Evaluated 54,613 ECG heartbeat images.
- **The Scope:** Tested 6 distinct architectures across 5 arrhythmia classes.



Two Real Challenges



1. Severe Class Imbalance

Most heartbeats are Normal. The **Fusion class is tiny**—about 85x smaller than Normal in the test set.

Because of this, plain accuracy is highly misleading. A model can look "accurate" just by guessing "Normal" every single time.



2. Domain Mismatch

Pre-trained deep learning models (like MobileNetV2) are tuned for **everyday photos** (cats, cars, people).

ECG line-plots are simple, stark black-and-white graphs. It wasn't obvious that features learned from natural photos would transfer well to clinical graphs.

The MIT-BIH Dataset Breakdown

Total: 54,613 ECG heartbeat images (37,178 for training, 17,435 for testing).

Class Label	Description	Training Images	Test Images
N	Normal beat	22,122	13,661
Q	Unknown / Paced beat	6,413	1,608
V	Ventricular ectopic beat	5,780	1,448
S	Supraventricular ectopic beat	2,222	557
F	Fusion beat (The Minority)	641	161

How Images Were Prepared

Two different tracks: the same underlying images, prepared differently for each model family.



Deep Learning Track

(Custom CNN, MobileNetV2)

- Resized strictly to **64×64 pixels**
- Maintained all **3 colour channels** (RGB)
- Scaled pixel values to a **[0, 1] range** for neural stability



Classical ML Track

(Random Forest, LogReg, SVM)

- Converted images to **grayscale**
- Resized to 64×64, then **flattened into 4,096 values**
- **Standardized** numerical scales (required for LogReg & SVM)

Six Models, Same Playing Field

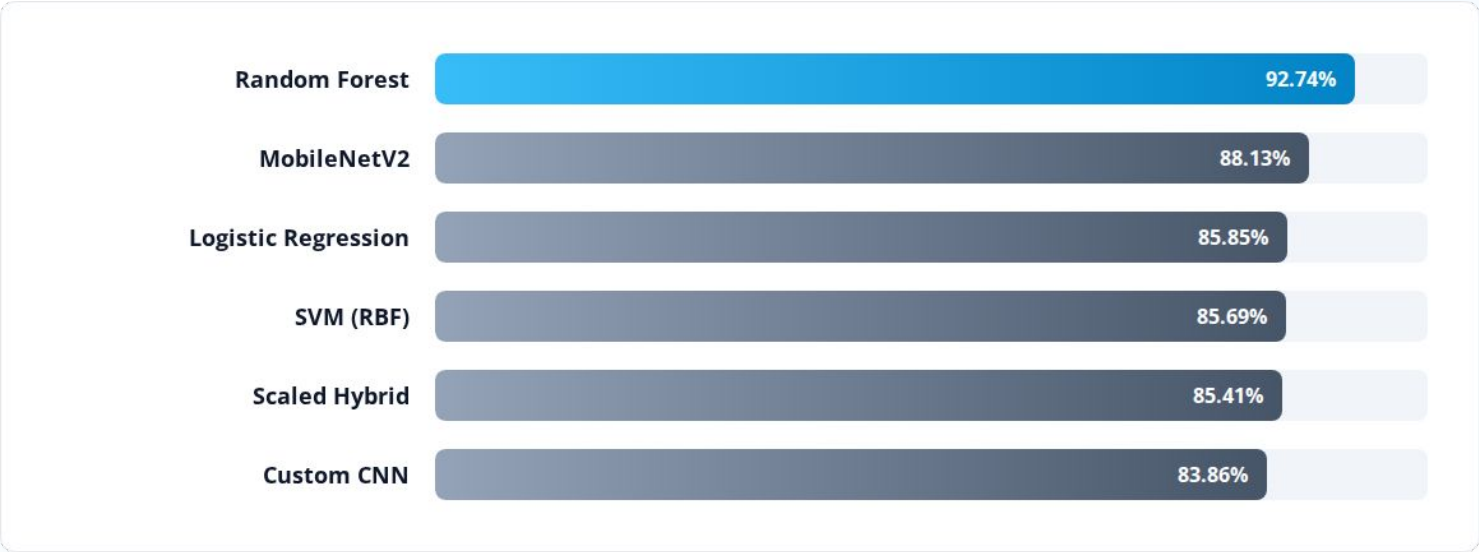
Deep Architectures

- > **Custom CNN:** A small, lightweight spatial network built entirely from scratch.
- > **MobileNetV2:** Used as a pretrained backbone to test deep Transfer Learning.
- > **Scaled Hybrid:** Deep features extracted by MobileNetV2, but classified by a classical SVM.

Classical Algorithms

- > **Random Forest:** An ensemble of 200 decision trees, utilizing class-balanced weights.
- > **SVM (RBF):** Support Vector Machine, trained on a representative 5,000-image subset.
- > **Logistic Regression:** A fast, simple linear baseline for flat arrays.

Overall Accuracy on the Test Set



Random Forest (a classical model) won on both raw accuracy AND Macro F1-Score (0.804).

The Struggle with the "Fusion" Class

0.27

MACRO F1-SCORE

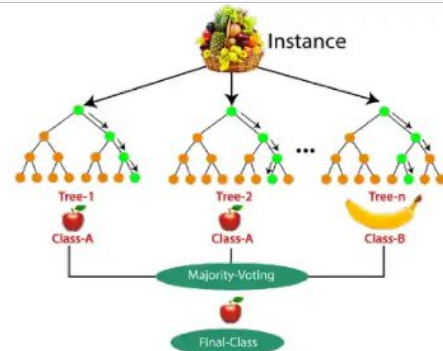
The Cost of Severe Imbalance

Every single model struggled significantly to identify the **Fusion (F)** beats. The number on the left (0.27) is the F1-score for the Fusion class on our *best* model (Random Forest).

With only 641 training images (compared to 22,000+ Normal images), the class is simply too small for algorithms to learn robust rules without synthetic data augmentation.

Why Did a Classical Model Win?

- > **Images are tightly cropped & aligned:** Every heartbeat plot has the same layout. Key features (like QRS peaks) land in the exact same pixel spots. Random Forest easily learns these fixed pixel rules. A CNN's ability to find shapes *anywhere* in an image is overkill here.
- > **Domain Mismatch:** MobileNetV2's weights are tuned for natural photos (textures, shading), which don't transfer well to harsh black-and-white lines.
- > **Training Setup Nuance:** Classical models utilized class-balanced weighting; the deep models in this run did not, giving classical ML an edge on minority classes.



Context & Future Scope

Comparison to Other Papers

Other papers convert ECG signals to images, but reach for heavier deep learning:

- **Huang et al. (2019):** Spectrograms + 2D CNN achieved **99.00%**
- **Rohmantri (2020):** 128x128 plots + deep CNN achieved **98.91%**
- **Our Result:** 64x64 plots + Random Forest achieved **92.74%**

Takeaway: SOTA is slightly higher, but requires massive GPU overhead. We proved lightweight models are highly competitive.

What We Left for Future Work

To keep the study honest, we deliberately scoped out certain techniques for future iteration:

- Applying **Focal Loss** to the CNN to actively penalize minority class errors during deep learning.
- Testing heavy architectures like ResNet or VGG directly on this specific dataset.
- Running **k-fold cross validation** to verify absolute model stability across splits.

The Final Verdict

For standardized, tightly cropped medical line-graphs, a well-tuned classical model (Random Forest) can beat deep transfer learning—while requiring a fraction of the computational power.